

Spurious symmetry-broken phase in a bidirectional two-lane ASEP with narrow entrances*

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As one of the paradigmatic models of non-equilibrium systems, the asymmetric simple exclusion process (ASEP) has been widely used to study many physical, chemical, and biological systems. The ASEP shows a range of nontrivial macroscopic phenomena, among which, the spontaneous symmetry breaking has gained a great deal of attention. Nevertheless, as a basic problem, it has been controversial whether there exist one or two symmetry-broken phases in the ASEP. Based on the mean field analysis and current minimization principle, this paper demonstrates that one of the broken-symmetry phases does not exist in a bidirectional two-lane ASEP with narrow entrances. Moreover, an exponential decay feature is observed, which has been used to predict the phase boundary in the thermodynamic limit. Our findings might be generalized to other ASEP models and thus deepen the understanding of the spontaneous symmetry breaking in non-equilibrium systems.

Keywords: Spontaneous symmetry breaking, mean field analysis, ASEP

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1. Introduction

Statistical investigation of non-equilibrium systems is a tough task. To understand the basic statistical properties of non-equilibrium systems, many paradigmatic models have been proposed. As one of the paradigmatic models of non-equilibrium systems, the asymmetric simple exclusion process (ASEP) has been widely used to study many physical, chemical, and biological systems,^[1–5] such as biopolymerization,^[6] gel electrophoresis,^[7] the kinetics of synthesis of proteins,^[8–10] biological transport,^[11–16] polymer dynamics in dense media,^[17] diffusion through membrane channels,^[18] surface growth,^[19,20] glassy dynamics,^[21,22] traffic flow,^[23–26] information flow,^[27–29] and so on.

The basic ASEP model is defined on a one-dimensional discrete lattice with L sites that are either occupied by a single particle or empty. Particles move along the lattice obeying a hard-core exclusion principle. This simple model can reproduce many non-equilibrium phenomena such as spontaneous symmetry breaking,^[30–42] boundary-induced^[43–45] and bulk-induced^[46] phase transitions, phase separation and condensation,^[47–51] shock formation,^[11,12,46,52] and so on. Thus, as Blythe and Evans^[4] pointed out, “our interest in the ASEP lies in its having acquired the status of a fundamental model of nonequilibrium statistical physics in its own right in much the same way that the Ising model has become a paradigm for equilibrium critical phenomena.”

The spontaneous symmetry breaking has been stud-

ied across various fields. The first ASEP model exhibiting spontaneous symmetry breaking is known as the “bridge model”.^[30,31] From then on, the phenomenon in ASEP models has gained a great deal of attention.^[32–42] However, the mechanism of this phenomenon is still not well understood. In particular, there are controversial reports about the number of symmetry-broken phases in the bridge model. The existence of one of the symmetry-broken phases (i.e., asymmetric low-density/low-density (LD/LD) phase) has been disputed on the basis of computer simulation by Arndt *et al.*^[42] Extensive high-precision Monte Carlo simulations indicated that the phase may disappear in the thermodynamic limit.^[32]

In the bridge model, particles moving in opposite directions interact with each other at every site of the lattice. Thus, Pronina and Kolomeisky^[33] argued that it is not clear how the interactions between these particles localized in the specific parts of the system will affect the symmetry breaking. Motivated by this fact, they have proposed to study the bidirectional two-lane asymmetric exclusion processes with narrow entrances, in which interactions of particles in the two channels only happen at the entrances.^[33] The system has been analyzed using a mean-field theoretical approach and extensive Monte Carlo computer simulations at different system sizes. It has been shown via mean field analysis that the model can also reproduce two phases with broken symmetry, i.e., the asymmetric high-density/low-density (HD/LD) phase, and the asymmetric LD/LD phase.

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To study the size-scaling dependence of the asymmetric LD/LD phase in their model, Pronina and Kolomeisky^[33] carried out computer simulations for the systems with different sizes up to $L = 1.2 \times 10^4$. It was shown that “the region of existence for the asymmetric LD/LD phase seems to shrink constantly with increasing L without reaching a saturation, which suggests that the phase probably does not exist in the thermodynamic limit”. However, Pronina and Kolomeisky^[33] also claimed that the numerical results are not very conclusive, thus more careful investigations of this phase are needed in order to understand the symmetry breaking phenomenon.

In this paper, we study the issue of whether the asymmetric LD/LD phase exists or not in the bidirectional two-lane ASEP with narrow entrances. We demonstrate that the asymmetric LD/LD phase should not exist based on the mean field analysis and the current minimization principle.

The paper is organized as follows. The model is briefly reviewed in Section 2. The mean field analysis is presented in Section 3. Section 4 summarizes the paper.

2. Model

The sketch of the model is shown in Fig. 1. The system consists of two parallel one-dimensional L lattices with particles moving in different lanes in the opposite directions. Each lattice site can be either empty, or occupied by one particle. Hopping between the lanes is not allowed. In the bulk, particles in lane 1 (2) hop to the right (left) with rate 1 if the next site is empty. At the entrance site, a particle is injected with rate α , provided that the site is empty and the exit site at the other lane is unoccupied. At the exit site, a particle is removed with rate β .

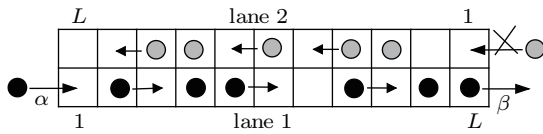


Fig. 1. Sketch of the model. Dark particles move along lane 1 from left to right, and gray particles move on lane 2 from right to left. Allowed transitions are shown by arrows. Crossed arrows indicate forbidden transitions.

3. Mean field analysis

We first present the results of simple mean field, which ignores correlations between any two sites. Then we present the cluster mean field results, in which the N -cluster mean field analysis and the current minimization principle demonstrate that the asymmetric LD/LD phase does not exist.

3.1. Simple mean field

Pronina and Kolomeisky^[33] presented a simple mean field analysis of the bidirectional two-lane system. They have shown that the necessary conditions for the existence of the

asymmetric LD/LD phase are

$$\frac{\alpha}{1 + \alpha + \alpha^2} < \beta < \frac{\alpha(1 - 2\alpha) + 2\alpha\sqrt{\alpha^2 - \alpha + 1}}{3}. \quad (1)$$

To determine whether the asymmetric LD/LD phase can exist or not, we resort to the current minimization principle proposed in Refs. [34], [35], and [46]. Specifically, for the parameter set that yields more than one solution, the solution that has the minimum system current is the most stable one, and thus exists. Other solutions do not exist.

When the necessary condition of the existence of the asymmetric LD/LD phase is satisfied, the system current, which is the sum of the currents on the two lanes and denoted as J_A , can be calculated

$$J_A = \left(\frac{\beta}{\alpha}\right)^2 + 2\beta - \frac{\beta}{\alpha}. \quad (2)$$

If the symmetric LD phase exists, then the system current, denoted as J_B , is

$$J_B = \frac{2\alpha^2\beta + \beta\sqrt{(\alpha + \beta)^2 - 4\alpha^2\beta} - \beta^2 - \alpha\beta}{\alpha^2}. \quad (3)$$

Finally, if the system is in an asymmetric HD/LD phase, then the system current, denoted as J_C , is

$$J_C = \beta(1 - \beta) + \alpha\beta(1 - \alpha\beta). \quad (4)$$

Figure 2 shows an example of the three currents at $\alpha = 0.4$. One can see that the curve of J_A intersects the curve of J_B at β_R and the curve of J_C at β_L . It can be easily proved that $\beta_L = \alpha/(1 + \alpha + \alpha^2)$ and $\beta_R = [\alpha(1 - 2\alpha) + 2\alpha\sqrt{\alpha^2 - \alpha + 1}]/3$. Thus, $J_A > J_C$ when β is in the range of Eq. (1). This means that even if the necessary condition of the existence of the asymmetric LD/LD phase is satisfied, the phase does not exist due to the current minimization principle. For a general value of α , the result does not change, which can be easily proved.

On the other hand, the curve of J_B intersects the curve of J_C at β_c . This means that when $\beta < \beta_c$, the system is in HD/LD phase. When $\beta > \beta_c$, the system is in symmetric LD phase.

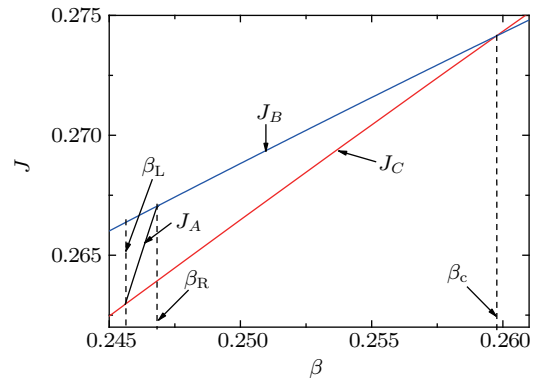


Fig. 2. (color online) The plot of J_A , J_B , and J_C versus β , obtained from simple mean field analysis. The parameter $\alpha = 0.4$.

3.2. Cluster mean field

While simple mean field analysis ignores correlations between any two sites, the correlations in the cluster have been considered in cluster mean field analysis. We number the sites from the entrance site to the exit site as site 1, 2, 3, ..., L , see Fig. 1. In the N -cluster mean field analysis, we consider N consecutive sites (from exit site) on one lane and N consecutive sites (from entrance site) on the other lane. Figure 3 shows the example with $N = 1$. In this case, there are four possible states of the system. We define P_{11} as a probability that the two sites are occupied, P_{00} as a probability that the two sites are empty, P_{10} , P_{01} as the probabilities that one site is empty and the other site is occupied, respectively.

Due to conservation of probabilities, one has

$$P_{00} + P_{01} + P_{10} + P_{11} = 1. \quad (5)$$

Now we can write the master equations of the evolution of the four probabilities.

$$\begin{aligned} \frac{dP_{00}}{dt} &= -(\alpha + p_1)P_{00} + \beta P_{01} + p_2 P_{10}, \\ \frac{dP_{01}}{dt} &= -\beta P_{01} + p_1 P_{00} + p_2 P_{11}, \\ \frac{dP_{10}}{dt} &= -(p_1 + p_2)P_{10} + \alpha P_{00} + \beta P_{11}, \\ \frac{dP_{11}}{dt} &= -(p_2 + \beta)P_{11} + p_1 P_{10}. \end{aligned}$$

Here p_1 is the hopping probability of a particle to the exit site of lane 1, and p_2 is the hopping probability of a particle at the entrance site of lane 2 (see Fig. 3).

In the stationary state, we have

$$\frac{dP_{00}}{dt} = \frac{dP_{01}}{dt} = \frac{dP_{10}}{dt} = \frac{dP_{11}}{dt} = 0.$$

Thus, one has

$$-(\alpha + p_1)P_{00} + \beta P_{01} + p_2 P_{10} = 0, \quad (6)$$

$$-\beta P_{01} + p_1 P_{00} + p_2 P_{11} = 0, \quad (7)$$

$$-(p_1 + p_2)P_{10} + \alpha P_{00} + \beta P_{11} = 0, \quad (8)$$

$$-(p_2 + \beta)P_{11} + p_1 P_{10} = 0.$$

However, note that only three of the four equations are independent.

Next we can analyze the asymmetric LD/LD phase. We express the bulk density of lane 2 as

$$\rho_2 = P_{10} + P_{11}. \quad (9)$$

Moreover, we assume

$$p_2 = 1 - \rho_2. \quad (10)$$

Thus, in the asymmetric LD/LD phase, we have seven variables P_{00} , P_{01} , P_{10} , P_{11} , p_1 , p_2 , ρ_2 and six equations (5)–(10). In order to solve the equations, p_1 needs to be given firstly. We cannot obtain an analytical result of the equations. Instead, we can obtain only a numerical result. When the equations are solved, we can calculate the bulk density of lane 1 from $J_1 = \rho_1(1 - \rho_1)$, and $J_1 = \rho_L \beta$, where $\rho_L = P_{11} + P_{01}$ is the density on the exit site of lane 1. We would like to mention that the two sites at the other end of the system satisfy similar equations.

Note that in Ref. [38], we introduce the bulk density of lane 1 as ρ_1 . Thus, the flow rate on lane 1 is $J_1 = \rho_1(1 - \rho_1)$, so that $\rho_L = \rho_1(1 - \rho_1)/\beta$. Based on $p_1(1 - \rho_L) = J_1$, we calculate $p_1 = \beta \rho_1(1 - \rho_1)/[\beta - \rho_1(1 - \rho_1)]$. However, a similar analytical relationship between p_1 and ρ_1 cannot be obtained in an N -cluster case ($N > 1$) due to the correlations in the cluster.

Next we consider the asymmetric HD/LD phase. Suppose lane 1 is in HD, one has $p_1 = 1 - \beta$. At the same time, Eqs. (5)–(10) are still satisfied. Thus, we have six variables P_{00} , P_{01} , P_{10} , P_{11} , p_2 , ρ_2 and six equations. Solving the equations, one can obtain the system current.

N -cluster mean field analysis with $N > 1$ can be carried out similarly and details will not be presented here. With the increase of N , the only assumption (Eq. (10)) in the cluster mean field analysis becomes more and more accurate, but the number of system states and accordingly the number of master equations increase as 2^{2N} .

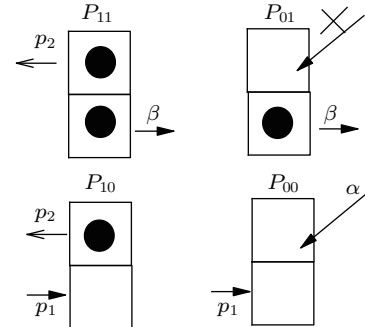


Fig. 3. Four possible states of the vertical cluster composed of the exit site of lane 1 and entrance site of lane 2 in one cluster mean field analysis.

3.2.1. One-cluster mean field analysis

Now we show the results of one-cluster mean field analysis. The black and red lines in Fig. 4 (top panels) show examples of the curve of $\rho_1 = f_1(\rho_2)$ and $\rho_1 = f_2(\rho_2)$, obtained from the two sites at two ends of the system. Here ρ_1 and ρ_2 are bulk densities on lanes 1 and 2, respectively. To illustrate the difference of the two curves more clearly, we show the difference of the two curves $f_1(\rho_2) - f_2(\rho_2)$ in the bottom panels. It can be seen that the two curves intersect at three points A, B, and C. Points B and C are symmetric about $\rho_1 = \rho_2$. Point A corresponds to $\rho_1 = \rho_2$. If one fixes α and decreases β ,

points B and C shift toward the two ends (Fig. 4(b)). If one increases β , points B and C shift toward point A until becoming in superposition with point A (Fig. 4(c)).

Thus, the necessary condition for the existence of an asymmetric LD/LD phase is that there should exist two points B and C such that $\rho_1 < \beta$ at point B and $\rho_2 < \beta$ at point C .

Note that in Ref. [35], the necessary condition has been incorrectly presented as $\rho_1 < 1/2$ at point B and $\rho_2 < 1/2$ at point C .

We have compared the three currents J_A , J_B , and J_C . The results are the same as in simple mean field analysis: the asymmetric LD/LD phase does not exist.

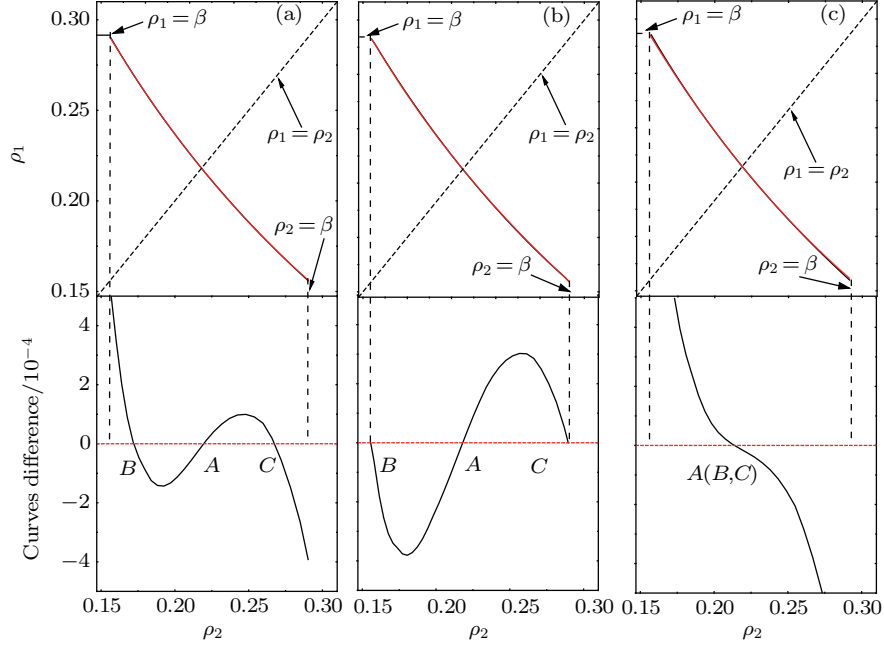


Fig. 4. (color online) The two curves of ρ_1 versus ρ_2 (top) and difference of the two curves (bottom). The parameter $\alpha = 0.8$, (a) $\beta = 0.3095$, (b) $\beta = 0.308436257$, (c) $\beta = 0.3106$.

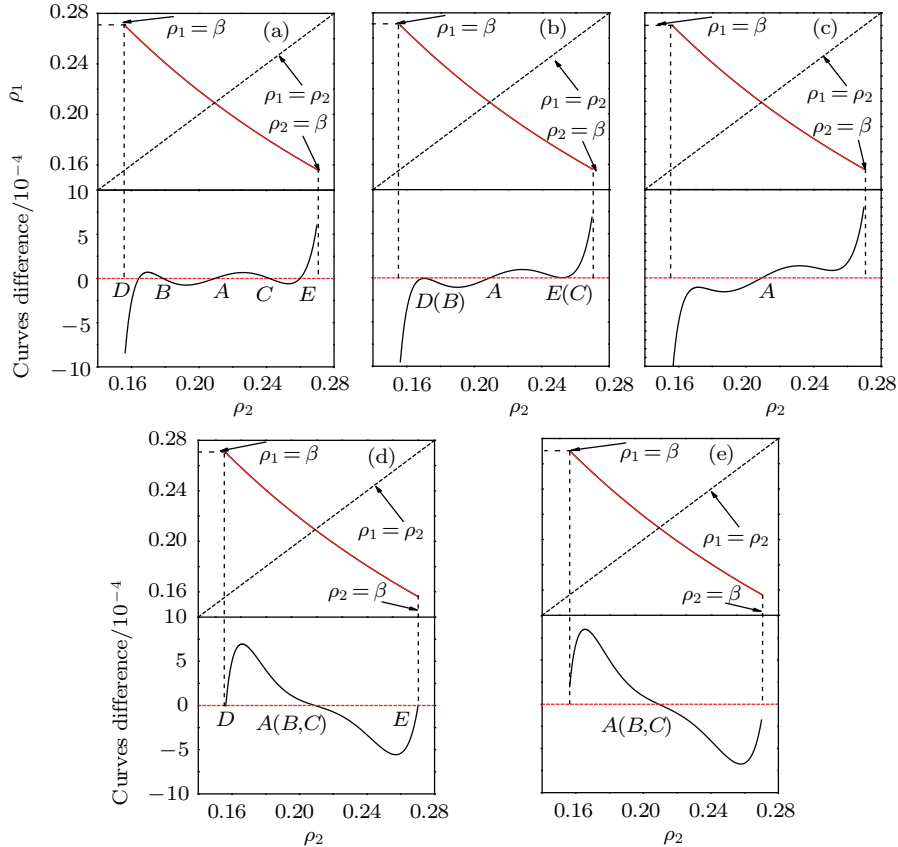


Fig. 5. (color online) The two curves of ρ_1 versus ρ_2 (top) and difference of the two curves (bottom). The parameter $\alpha = 0.8$, (a) $\beta = 0.27014$, (b) $\beta = 0.270123$, (c) $\beta = 0.2701$, (d) $\beta = 0.2702699$, (e) $\beta = 0.2703$.

3.2.2. N -cluster mean field

Next we perform N -cluster mean field analysis and increase N from 1. The result does not change until $N = 4$. However, when $N > 4$, the results become different. Figure 5 shows examples of the curves of ρ_1 versus ρ_2 and $f_1(\rho_2) - f_2(\rho_2)$ in the 5-cluster mean field analysis. One can see that the two curves intersect at five points A–E. Points B and C are still symmetric about $\rho_1 = \rho_2$. Point A still corresponds to $\rho_1 = \rho_2$. Points D and E are also symmetric about $\rho_1 = \rho_2$. This means that there are three solutions: points B and C correspond to one asymmetric LD/LD solution I; points D and E correspond to the other asymmetric LD/LD solution II; and point A still corresponds to a symmetric LD solution. If one fixes α and decreases β , points B and C shift toward two ends until coinciding with D and E, respectively, see Fig. 5(b). With the further decrease of β , the asymmetric LD/LD solution vanishes, see Fig. 5(c). If one increases β , points B and C shift toward point A until becoming in superposition with point A, see Fig. 5(d). With the further increase of β , the asymmetric LD/LD solution vanishes too, see Fig. 5(e).

We compare the flow rates of asymmetric HD/LD solution, asymmetric LD/LD solution I, asymmetric LD/LD solution II and symmetric LD solution, see Fig. 6. In the range of asymmetric LD/LD solution $\beta_L < \beta < \beta_R$, in which the necessary condition of the existence of an asymmetric LD/LD phase is satisfied, the flow rates of two asymmetric LD/LD solutions are both larger than that of an asymmetric HD/LD solution. Changing the value of α , the result does not change, either.

We have increased N until $N = 6$, the result does not change anymore. Although we are not able to prove rigorously, the mean field analysis and the current minimization principle strongly demonstrate that the asymmetric LD/LD

phase does not exist.

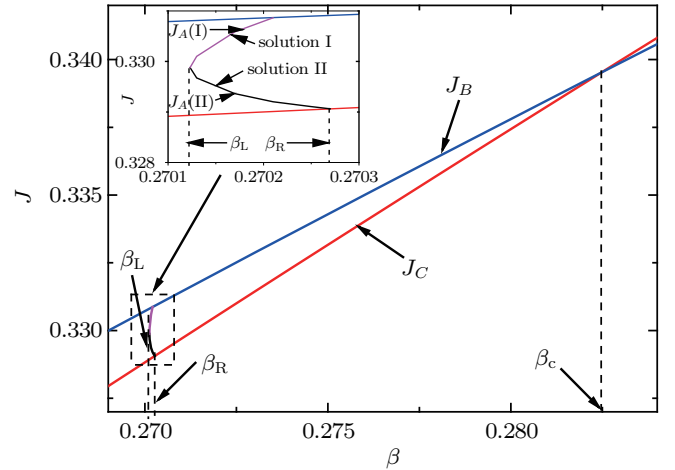


Fig. 6. (color online) The plot of J_A , J_B , and J_C versus β , obtained from 5-cluster mean field analysis. The parameter $\alpha = 0.8$.

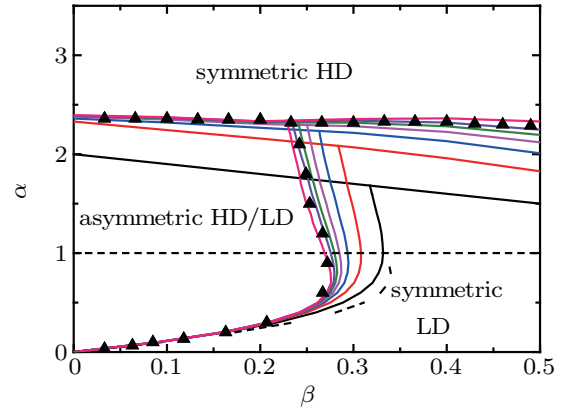


Fig. 7. (color online) Phase diagram of the system. Scattered data are from simulations (system size $L = 10^4$). Dash lines are from simple mean field analysis. The black line, red line, blue line, magenta line, olive line, and violet line correspond to N -cluster mean field analysis with $N = 1-6$, respectively. Pink lines are predicted results in the limit $N \rightarrow \infty$ based on exponential decay feature.

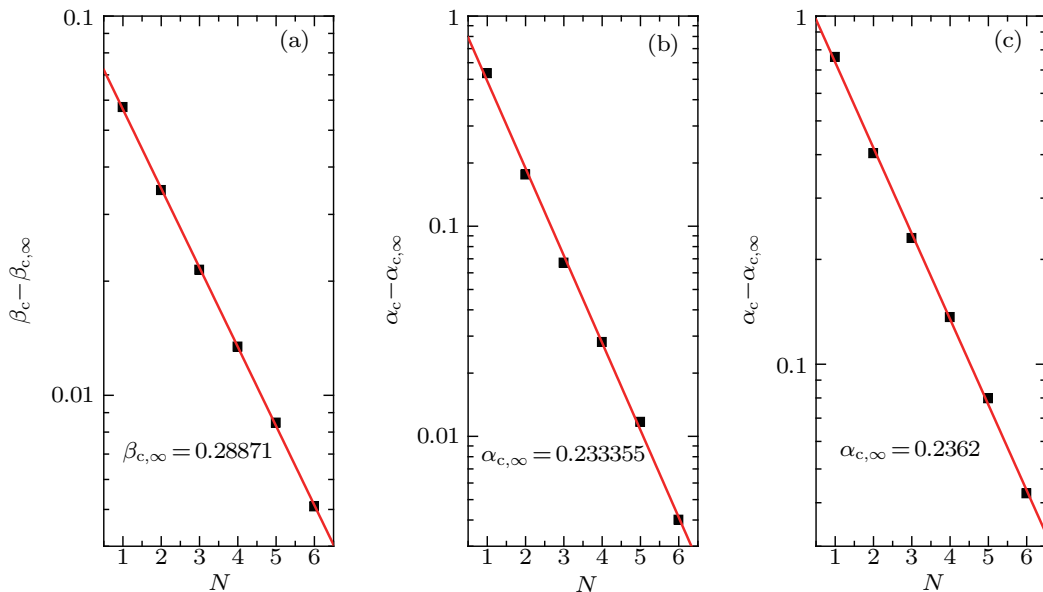


Fig. 8. (color online) Dependence of the boundary on N in N -cluster mean field analysis. (a) $\alpha = 0.9$, (b) $\beta = 0.2$, (c) $\beta = 0.4$. Scattered points are N -cluster mean field results, the red lines show the fit of exponential decay.

Figure 7 shows the boundaries obtained from simple mean field, cluster mean field analysis, and simulations. Note that since the asymmetric LD/LD phase still exists in the simulations, we show the boundary separating the asymmetric HD/LD phase from the asymmetric LD/LD phase. The boundary separating the asymmetric LD/LD phase from the symmetric LD phase is not shown in the simulation results. One can see that cluster mean field analysis performs better than simple mean field analysis. With the increase of N , the analytical boundaries approach the simulation ones.

Finally we study the dependence of phase boundary on N in N -cluster mean field analysis. Figure 8(a) shows that at a given α , the boundary separating the asymmetric HD/LD phase from the symmetric LD phase is remarkably consistent with the exponential decay $\beta_c - \beta_{c,\infty} \propto \exp(aN + b)$. Therefore, one expects that $\beta_c = \beta_{c,\infty}$ when $N \rightarrow \infty$. Figures 8(b) and 8(c) show that the boundary separating symmetric HD phase from the other two phases also exhibit the exponential decay feature. The predicted boundaries in the limit $N \rightarrow \infty$ based on the exponential decay feature are also shown in Fig. 7. Simulations in larger system size need to be carried out to validate the prediction. However, one needs to be very careful with the simulations in very large systems due to the fact that the random-number generator produces a pseudorandom number series instead of a true random number series.^[53]

4. Conclusion and perspectives

To summarize, this paper studies symmetry breaking phenomenon in a bidirectional two-lane ASEP with narrow entrances, in which it is controversial whether an asymmetric LD/LD phase exists or not. We demonstrate that the asymmetric LD/LD phase should not exist based on mean field analysis and current minimization principle. The cluster mean field analysis performs better than simple mean field analysis. With the increase of N -cluster, the analytical boundaries approach the simulation ones. We observe an exponential decay feature in N -cluster mean field analysis, which has been used to predict the phase boundary in the thermodynamic limit.

The findings can be important for the understanding of the symmetry breaking phenomenon in non-equilibrium systems. The method adopted here might be generalized to other ASEP models. The results can also shed some light on the dynamics of pedestrian flow, traffic flow and biological transport.

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